# MSME Ledger Digitisation

## Problem statement, objectives and scope

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**The project builds a system that turns photographs and scans of the handwritten ledgers kept by small businesses into editable digital tables, which the owner checks, corrects, saves and exports.**

## 1. Problem statement

Many micro, small and medium enterprises (MSMEs) in Zimbabwe keep their business records by hand: sales books, stock sheets, debtor lists, receipt books and cash books. These paper records cause practical problems for the businesses that keep them:

- **Hard to use.** Totals, searches and comparisons over weeks or months must be worked out by hand.
- **Hard to share.** Owners cannot easily give a bank, microfinance lender or supplier a clear record of their sales and cash flow.
- **Easily lost.** Exercise books are damaged, misplaced or worn out, and there is no backup.
- **Slow and error-prone to digitise by hand.** Retyping pages into a spreadsheet takes time that small business owners rarely have, and introduces typing errors.

General-purpose text-recognition (OCR) tools do not solve this well. They are built for printed text, while these ledgers are handwritten, often faded, skewed or photographed in poor light, written partly in Shona or Ndebele, full of ditto marks, and laid out differently by every business. A tool that assumes one fixed layout cannot read a stock sheet, a debtor list and a receipt book alike.

> **Problem statement.** MSME owners lack a low-cost, easy-to-use tool that converts photographs of their handwritten ledgers, whatever their layout, into accurate, editable digital records that they can check, correct, keep securely and export.

## 2. Aim and objectives

**Aim:** to design, build and evaluate a system that converts photographs and scans of handwritten MSME ledgers into editable, exportable digital tables, with the business owner checking every result.

### Objectives

1. **Prepare the images.** Develop an image preprocessing pipeline (straightening, noise removal, contrast enhancement and binarisation) that makes phone-captured ledger pages easier to read.
2. **Compare OCR engines.** Implement two open-source engines, Tesseract and EasyOCR, and measure their accuracy against hand-typed ground truth using character and word error rates (CER and WER).
3. **Combine the engines.** Design a rule-based fusion and field-extraction method that uses each engine's strengths to produce structured rows (date, item, quantity, price, total) with a lower error rate than either engine alone.

4. **Read any layout.** Integrate a vision-language model (Claude) that reads each document's own column and row headings, so different kinds of records can be digitised, with the open-source pipeline as a fallback.
5. **Let owners check and correct.** Build a mobile-friendly web application, with a companion Android app, where owners upload pages, see uncertain cells highlighted, correct mistakes and save their work in a private account.
6. **Get the data out.** Provide exports in Excel (a sheet per page), CSV and JSON, and a bank-statement layout, so records can be kept, analysed and shared with lenders.
7. **Evaluate the system.** Measure recognition accuracy, processing time and cost per page, online and offline, and report the results.

# 3. Scope

The table sets out what the project covers. Section 3.2 lists what it deliberately leaves out.

## 3.1 In scope

| Area | Covered |
|---|---|
| **Users** | Owners and staff of Zimbabwean MSMEs who keep handwritten records. |
| **Records** | Sales books, stock sheets, debtor lists, receipt books, cash books and similar tables, in any layout. |
| **Input** | Phone photos (JPEG, PNG, TIFF, BMP, WebP) and scanned PDFs: up to 40 pages and 100 MB per file. |
| **Language** | Records in English with Shona and Ndebele words; the interface in English, chiShona and isiNdebele. |
| **Reading** | Online: Claude vision reads each page's own headings. Offline or as a fallback: Tesseract and EasyOCR, fused by rules. Ditto marks and letter-O zeros are tidied automatically. |
| **Checking** | The owner reviews every result before saving. Unreadable or uncertain cells are shaded amber, beside the photo of the page. |
| **Platforms** | Web application (Python, Flask) used in a phone or computer browser, hosted online on Hugging Face Spaces or offline on a laptop over local Wi-Fi; an Android app (Flutter). |
| **Data** | Individual accounts with hashed passwords; each user sees only their own records; SQLite storage with a private backup. |
| **Output** | Excel, CSV and JSON downloads, and a bank-statement layout. |
| **Evaluation** | Recognition and field-extraction accuracy (CER, WER, cells correct), time and cost per page. |

### 3.2 Out of scope

- **Training a new handwriting model.** The project uses existing OCR engines and a hosted model; collecting and labelling enough handwriting to train one was not feasible in the time available.
- **Accounting.** Bookkeeping, tax returns and financial statements are not produced; the system digitises records, it does not account for them.
- **Business outcomes.** Effects on profitability, lending decisions or loan repayment are not measured; evaluation covers the system's accuracy, speed and cost.
- **Integrations.** No direct links to banks, mobile money, or accounting software; data leaves the system as downloaded files.
- **Unchecked automation.** Results are not guaranteed correct without the owner's review, and the offline reader uses fixed columns rather than each document's own headings.
- **iPhone app.** The phone app is built for Android; iPhone users use the website.

> **Changes from the original proposal**
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> The proposal limited the system to JPEG and PNG photos, a web application with no cloud deployment, and data kept on the host machine. The finished system also accepts scanned PDFs, is hosted online (with an offline laptop mode), uses Claude to read varied layouts, and keeps each user's records in a private account. Chapter 1's scope and the security section (4.4.3) should be updated to match.